

Geometry

Երկրաչափություն

COURSE CODE	MTH-GE
PROGRAM	High School Mathematics · Ավագ դպրոցի մաթեմատիկա
LEVEL	Typically Grade 9 or 10
DURATION	108–126 instructional hours (approx. 90 lessons)
PREREQUISITE	Algebra I (MTH-A1) or equivalent: linear equations, systems, the coordinate plane, radicals.
LEADS TO	Algebra II (MTH-A2)

COURSE DESCRIPTION

Դասընթացի նկարագրություն

Geometry is the course in which a student learns what it means to prove something. Building on the transformational foundation laid in Grade 8, congruence and similarity are defined through rigid motions and dilations rather than asserted, and the standard triangle criteria are established as consequences rather than accepted as axioms. Students develop formal proof in several registers, study circles, right triangle trigonometry, coordinate geometry, three-dimensional measurement, and geometric probability. The course is as much a course in reasoning as in shape.

TEACHING APPROACH

Դասավանդման մոտեցումը Վարժարանում

Վարժարան teaches Geometry as an argument, not as a catalogue of formulas. The proof is the point. That said, the two-column proof is a format, not a goal — students write proofs in paragraph, flow-chart, two-column, and coordinate form, choosing the one that best expresses the argument, because a student who can only produce two-column proofs has learned a template rather than a way of thinking. Teachers are asked to require a reason for every claim from the very first lesson, including in computational problems, so that justification is normal rather than a special activity reserved for proof days. Construction with compass and straightedge is retained deliberately: it is where students discover the theorems they will later prove.

PLACEMENT AND READINESS INDICATORS

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The placement assessment checks each indicator below. A student missing two or more is enrolled with a remediation plan attached to the personalized curriculum.

READINESS INDICATOR	WHAT IT TELLS THE TEACHER
Solves multi-step linear equations and systems accurately	Nearly every geometric computation reduces to an algebraic one; weakness compounds.
Simplifies radical expressions	Required from the Pythagorean theorem onward and throughout trigonometry.
Uses the coordinate plane and computes slope and distance	Prerequisite for the coordinate geometry unit and for analytic proof.
Describes a transformation of a figure on the coordinate plane	The transformational definitions of congruence and similarity depend on it.

COURSE LEARNING OUTCOMES

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On completion of this course the student will be able to:

- 1 Know precise definitions of angle, circle, perpendicular line, parallel line, and line segment based on undefined notions.
- 2 Represent and describe transformations in the plane and determine those that preserve distance and angle.
- 3 Understand congruence in terms of rigid motions and use it to establish the triangle congruence criteria.
- 4 Prove theorems about lines, angles, triangles, and parallelograms.
- 5 Make formal geometric constructions with a variety of tools and methods.
- 6 Understand similarity in terms of similarity transformations and establish the AA criterion.
- 7 Prove theorems involving similarity, including the Pythagorean theorem via similar triangles.
- 8 Define trigonometric ratios for acute angles and use them to solve right triangles.
- 9 Apply the Law of Sines and the Law of Cosines to find unknown measures in any triangle.
- 10 Understand and apply theorems about circles, including inscribed angles, chords, tangents, and secants.
- 11 Find arc lengths and areas of sectors, and derive the equation of a circle.
- 12 Use coordinates to prove simple geometric theorems algebraically and to compute perimeters and areas.
- 13 Explain volume formulas and use them to solve problems, including with density.
- 14 Apply geometric concepts in modelling situations and identify cross-sections and solids of revolution.
- 15 Understand independence and conditional probability and use the rules of probability.

COURSE UNITS

Դասընթացի բաժիններ

Each unit below lists the lesson-level topics a teacher delivers, the objectives that define success, the observable mastery criteria used to update the student's progress profile, and the misconceptions this unit reliably produces together with the recommended teacher response.

UNIT 1 • 12-13 LESSONS

Foundations, Constructions, and Proof

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TOPICS COVERED

- Undefined terms: point, line, plane, distance along a line, distance around a circular arc
- Precise definitions of segment, ray, angle, circle, perpendicular, and parallel
- Segment and angle addition; midpoints and bisectors
- Compass and straightedge constructions: copying a segment and an angle
- Constructing a perpendicular bisector, a perpendicular through a point, an angle bisector
- Constructing parallel lines; inscribing regular polygons in a circle
- The structure of a deductive argument: given, statement, reason, conclusion
- Conditional statements, converse, inverse, contrapositive
- Inductive versus deductive reasoning; the role of a counterexample
- Writing proofs in paragraph, two-column, and flow-chart form
- Proving angle relationships: vertical angles, linear pairs
- Parallel lines cut by a transversal and the resulting angle theorems
- Proving lines parallel from angle relationships

LEARNING OBJECTIVES

- 1 State a precise definition of each basic geometric object and explain why the precision matters.
- 2 Perform each core construction and justify why it produces what it claims to.
- 3 Write a valid proof of an angle relationship in at least two different formats.
- 4 Distinguish a conditional statement from its converse and evaluate the truth of each.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Performs all core constructions accurately in 9 of 10 trials.
- Produces a valid, complete proof of a parallel-line angle theorem.
- Correctly forms and evaluates converse and contrapositive statements in 9 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
A diagram is treated as evidence: two segments that look equal are assumed equal.	Only marked or derived information may be used. Present a deliberately misleading diagram early in the unit — the lesson is learned in one experience.
The converse of a true statement is assumed true.	Produce a counterexample together. This is the logical error that most often invalidates student proofs.

MISCONCEPTION	RECOMMENDED RESPONSE
Proof is treated as a memorized sequence rather than an argument.	Ask the student to explain the proof aloud without notation. If they cannot narrate it, they have not understood it.

TEACHING NOTE

Constructions are not a historical curiosity. A student who constructs a perpendicular bisector and then observes that every point on it is equidistant from the endpoints has discovered a theorem, and will prove it with conviction rather than compliance.

UNIT 2 • 11-12 LESSONS**Transformations and Congruence**

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TOPICS COVERED

- Transformations as functions on the plane
- Translations, reflections, and rotations described precisely
- Rigid motions and the properties they preserve
- Composition of transformations
- Symmetry: mapping a figure onto itself
- Congruence defined by rigid motion
- Establishing the ASA, SAS, and SSS criteria from the definition
- Why SSA does not determine a triangle
- HL for right triangles
- CPCTC and its use in proofs
- Proving theorems about triangles: isosceles triangle theorem and its converse
- The triangle midsegment theorem
- Points of concurrency: centroid, circumcenter, incenter, orthocenter
- Proving theorems about parallelograms and their special cases
- The triangle inequality and hinge theorem

LEARNING OBJECTIVES

- 1 Describe a specific sequence of rigid motions mapping one figure onto another.
- 2 Explain how the SAS criterion follows from the definition of congruence via rigid motions.
- 3 Prove a theorem about triangles or parallelograms using congruent triangles and CPCTC.
- 4 Explain, with a construction, why SSA fails to determine a triangle.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Produces a valid transformation sequence for a congruence in 9 of 10 trials.
- Writes complete triangle congruence proofs in 4 of 5 tasks.
- Correctly identifies which criterion applies, or that none does, in 9 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
SSA is used as a congruence criterion.	Construct the ambiguous case physically with a compass and let the student see two distinct triangles from the same data.
Corresponding parts are matched incorrectly because the congruence statement order is ignored.	The order of the letters in $\triangle ABC \cong \triangle DEF$ is information. Require correspondence to be written before any conclusion is drawn.
CPCTC is invoked before congruence has been established.	Congruence first, then corresponding parts. The order is the logic; reversing it is circular.

TEACHING NOTE

Deriving the congruence criteria from rigid motions rather than assuming them is the modern approach and it is worth the extra time. It converts three memorized abbreviations into three theorems the student can defend.

UNIT 3 • 13-14 LESSONS**Similarity and Right Triangle Trigonometry**

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TOPICS COVERED

- Dilations, centre, and scale factor
- Properties of dilations and what they preserve
- Similarity defined by a similarity transformation
- Establishing the AA criterion; SSS and SAS similarity
- Proportional reasoning in similar figures
- The side-splitter theorem and its converse
- Proving the Pythagorean theorem using similar triangles
- Geometric mean relationships in right triangles
- Special right triangles: 45-45-90 and 30-60-90
- Defining sine, cosine, and tangent as ratios in similar right triangles
- Why the ratios depend only on the angle
- Solving right triangles for sides and angles
- Inverse trigonometric functions for angle measure
- Angles of elevation and depression; applied problems
- The sine relationship for the area of a triangle
- The Law of Sines and the Law of Cosines with derivations
- The ambiguous case of the Law of Sines

LEARNING OBJECTIVES

- 1 Explain why the trigonometric ratios are well defined, using similarity.
- 2 Prove the Pythagorean theorem using similar triangles.
- 3 Solve any triangle using the Law of Sines or Cosines and choose correctly between them.
- 4 Model and solve an applied problem involving angles of elevation or depression.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Solves right triangles at 92% accuracy including inverse trigonometric work.
- Chooses correctly between the Law of Sines and Cosines in 9 of 10 trials.
- Produces a valid similar-triangle proof of the Pythagorean theorem.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The trigonometric ratios are memorized without understanding why they depend only on the angle.	Draw three similar right triangles of different sizes and compute the ratios in all three. The invariance must be observed.
The opposite and adjacent sides are identified relative to the wrong angle.	Mark the reference angle first and label all three sides relative to it before writing any ratio.
The calculator is in radian mode.	Check the mode at the start of every trigonometry lesson. This costs five seconds and prevents an entire class of unexplained errors.

TEACHING NOTE

The fact that the trigonometric ratios are well defined because of similarity is the intellectual link between the two halves of this unit, and it is frequently left implicit. Make it explicit — it is the reason trigonometry works at all.

UNIT 4 • 10-11 LESSONS**Circles**

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TOPICS COVERED

- Circle vocabulary: centre, radius, chord, diameter, secant, tangent, arc, sector, segment
- Central angles and arc measure
- The inscribed angle theorem and its proof
- Angles inscribed in a semicircle
- Inscribed quadrilaterals in a circle
- Chord properties and perpendicular bisectors of chords
- Tangent lines and the radius-tangent perpendicularity theorem
- Two tangents from an external point
- Angles formed by chords, secants, and tangents
- Segment length relationships: chord-chord, secant-secant, secant-tangent
- Arc length as a proportion of the circumference
- Sector area as a proportion of the circle area
- Radian measure introduced as arc length over radius
- Constructing inscribed and circumscribed circles of a triangle
- The equation of a circle derived from the distance formula
- Completing the square to find the centre and radius from a general equation

LEARNING OBJECTIVES

- 1 Prove the inscribed angle theorem for at least one configuration.
- 2 Apply chord, tangent, and secant relationships to solve for unknown lengths and angles.
- 3 Compute arc length and sector area and express an angle in radians.
- 4 Derive the equation of a circle and recover its centre and radius by completing the square.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Applies circle angle and segment theorems at 88% accuracy.
- Computes arc length and sector area correctly in 9 of 10 trials with units.
- Converts a general circle equation to centre-radius form in 9 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Inscribed angle and central angle measures are interchanged.	The inscribed angle is half the central angle on the same arc. Have the student mark the arc first, then identify the vertex position.
Arc length and arc measure are confused.	Arc measure is in degrees and is independent of size; arc length is a distance and is not. Compare two concentric arcs with the same measure.
Completing the square is performed without balancing the equation.	Whatever is added inside must be added on the other side. Write the addition on both sides explicitly.

TEACHING NOTE

Radian measure is introduced here rather than deferred entirely to Precalculus. Presenting it as arc length divided by radius — a ratio, therefore a pure number — makes it far less mysterious when it returns as the standard unit.

UNIT 5 • 8-9 LESSONS

Coordinate Geometry

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TOPICS COVERED

- The distance formula derived from the Pythagorean theorem
- The midpoint formula and partitioning a segment in a given ratio
- Slope criteria for parallel and perpendicular lines
- Writing equations of parallel and perpendicular lines through a point
- Proving a figure is a specific quadrilateral using coordinates
- Proving a triangle is isosceles, right, or equilateral using coordinates
- Proving a point lies on a perpendicular bisector
- Computing perimeter and area of polygons in the coordinate plane
- The equation of a circle and its intersections with lines
- Choosing coordinates strategically to simplify a proof
- Comparing synthetic and analytic proofs of the same theorem

LEARNING OBJECTIVES

- 1 Prove a geometric claim about a figure using coordinates and algebra.
- 2 Choose a coordinate placement that simplifies a general proof.
- 3 Find the point partitioning a segment in a given ratio.
- 4 Compare a coordinate proof and a synthetic proof of the same result and evaluate their relative merits.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Produces valid coordinate proofs in 4 of 5 tasks.
- Computes distance, midpoint, and partition points at 92% accuracy.
- Selects an efficient coordinate placement for a general proof in 4 of 5 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
A specific numerical example is offered as a proof of a general claim.	One case is not a proof. For a general claim, use variables for the coordinates.
Coordinates are placed arbitrarily, producing unnecessarily messy algebra.	Place a vertex at the origin and a side along an axis. Teach this as a strategic decision, not a convention.

TEACHING NOTE

The comparison between synthetic and analytic proofs of the same theorem is one of the most illuminating exercises in the course. It shows the student that mathematics offers genuine choices of method, not one prescribed path.

UNIT 6 • 8-9 LESSONS

Three-Dimensional Geometry and Modelling

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TOPICS COVERED

- Cross-sections of three-dimensional figures
- Solids of revolution generated by rotating a two-dimensional figure
- Cavalieri's principle and its use in justifying volume formulas
- Volume of prisms, cylinders, pyramids, cones, and spheres
- Informal arguments for the cone and sphere volume formulas
- Surface area of prisms, cylinders, pyramids, cones, and spheres
- The effect of scaling on length, area, and volume
- Density problems: population density, mass per unit volume
- Modelling real objects with geometric solids
- Design problems with physical and cost constraints

LEARNING OBJECTIVES

- 1 Identify the cross-section of a solid cut by a specified plane.
- 2 Use Cavalieri's principle to justify a volume formula.

- 3 Solve a density or design problem using geometric modelling.
- 4 Explain how length, area, and volume scale under a dilation, with justification.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Computes volumes and surface areas at 92% accuracy with correct units.
- Identifies cross-sections correctly in 8 of 10 trials.
- Produces a defensible geometric model with stated assumptions for a real design problem.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Scaling relationships for area and volume are assumed linear.	Under a scale factor k , area scales by k^2 and volume by k^3 . Build the models and count.
Surface area is computed with the slant height and volume with the vertical height, or the two are interchanged.	Label both heights on every diagram before using either. Most cone errors are labelling errors.

TEACHING NOTE

Modelling problems in this unit should be genuinely open: give the student a real object and a question, and require them to state their simplifying assumptions. Stating assumptions explicitly is a mathematical skill and is rarely taught anywhere else.

UNIT 7 · 7-8 LESSONS

Probability

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TOPICS COVERED

- Sample spaces and events as subsets
- Unions, intersections, and complements
- Independent events and the multiplication rule
- Conditional probability and its interpretation
- Two-way tables and conditional probability
- Testing for independence using conditional probability
- The addition rule for events that may overlap
- The general multiplication rule
- Permutations and combinations and when each applies
- Using counting techniques to compute probabilities
- Geometric probability based on length and area
- Using probability to evaluate a decision

LEARNING OBJECTIVES

- 1 Compute a conditional probability from a two-way table and interpret it in context.
- 2 Determine whether two events are independent using the definition, not intuition.
- 3 Choose correctly between a permutation and a combination and justify the choice.

- 4 Use expected value informally to evaluate a decision.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Computes conditional probabilities correctly in 9 of 10 trials.
- Tests independence correctly in 8 of 10 trials with justification.
- Distinguishes permutation from combination situations in 9 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Independence is confused with mutual exclusivity.	Mutually exclusive events with non-zero probability are never independent. Construct an example of each and compare directly.
Conditional probability is computed with the wrong denominator.	$P(A B)$ restricts attention to B. Shade the reduced sample space on the table before computing.
A combination is used where order matters.	Ask: would swapping two selections give a different outcome? If yes, order matters.

TEACHING NOTE

Conditional probability is where most students' intuition is reliably wrong, which makes it excellent teaching material. Use a well-chosen counterintuitive example and let the student's surprise carry the lesson.

ASSESSMENT AND MASTERY POLICY

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ASSESSMENT	WHEN	WHAT IT MEASURES
Placement Assessment	Before enrollment	Algebra I retention, radicals, coordinate plane, and transformations.
Unit Check	End of each unit · 55 min	Unit mastery criteria including at least one full proof.
Proof Portfolio	Continuous	A curated set of the student's best proofs across all formats, reviewed at each unit boundary; the primary evidence of reasoning growth.
Construction Practical	Units 1 and 3	Compass-and-straightedge constructions performed live with justification.
Mid-Year Assessment	After Unit 3	Cumulative Units 1–3; weighted toward congruence, similarity, and proof.
End-of-Course Assessment	After Unit 7	All course outcomes in a format matching state end-of-course examinations.

Վարժարանը reports mastery, not grades. Each topic in the student's profile is marked **Mastered**, **Developing**, or **Needs Review** against the criteria stated in each unit above. A topic is marked Mastered only when the criterion is met on two separate occasions at least one week apart — this prevents short-term recall from being recorded as durable learning. Topics marked Needs Review are automatically re-entered into homework and into the opening minutes of subsequent lessons until they are re-mastered.

PACING OPTIONS

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TRACK	CADENCE	EXPECTED COMPLETION
Standard Year	2 lessons/week, 60 min	Full course in approximately 45 weeks.
Extended	1 lesson/week, 75 min	Full course across two academic years.
Accelerated	3 lessons/week	Full course in approximately 30 weeks.
Support / Co-Seated	2 lessons/week alongside school	Paced to the student's school syllabus.
Proof Intensive	2 lessons/week, 8 weeks	Units 1 and 2 only — for students whose school course omitted formal proof.

Pacing is assigned after the placement assessment and reviewed at the mid-year assessment. A student who is meeting mastery criteria early may be moved to the accelerated track; a student who is not may be moved to intensive without any change in the course content itself.

HOMEWORK AND INDEPENDENT PRACTICE

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Fifty to sixty minutes per lesson. Every assignment contains at least one proof and at least one problem requiring a construction or an accurate diagram. Computational problems must state the theorem or definition justifying each step — a numerical answer without a reason is marked incomplete, including in the earliest units. Cumulative review is drawn from all prior units, with particular attention to Algebra I skills, which decay quickly during a geometry year and are needed intact for Algebra II.

PROGRESS MILESTONES REPORTED TO PARENTS

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These are the achievements the parent portal announces as the student reaches them. They replace attendance counts as the primary measure of progress.

- ◆ States precise definitions of the basic geometric objects.
- ◆ Performs and justifies all core compass-and-straightedge constructions.
- ◆ Writes valid proofs in paragraph, two-column, and flow-chart formats.
- ◆ Derives the triangle congruence criteria from rigid motions.
- ◆ Proves theorems about triangles, parallelograms, and parallel lines.
- ◆ Establishes similarity via transformations and proves the Pythagorean theorem by similar triangles.
- ◆ Solves right triangles with trigonometric ratios and inverse functions.
- ◆ Applies the Law of Sines and the Law of Cosines, including the ambiguous case.
- ◆ Proves and applies the circle theorems; derives the equation of a circle.

- ◆ Produces valid coordinate proofs and chooses coordinates strategically.
- ◆ Justifies volume formulas and solves density and design problems.
- ◆ Computes conditional probability and tests independence.
- ◆ Completed Geometry — ready for Algebra II.

MATERIALS AND RESOURCES

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- Compass, straightedge, protractor, and ruler — physical set at home is required, not optional.
- Dynamic geometry software (GeoGebra or equivalent) for exploration and conjecture.
- Scientific or graphing calculator with trigonometric functions.
- Physical solids or nets for the three-dimensional unit.
- Վարժարան Geometry problem bank — approximately 1,800 items including 300 proof tasks graded by difficulty.

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